

Network Monitoring — Guided Notes

Unit 3.1.2 | CompTIA Network+ N10-009 Obj. 3.1 | N10-008 3.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

SNMP (Simple Network Management Protocol) uses a _____ model where the monitoring server polls devices, and a _____ model where devices send unsolicited alerts when a threshold is crossed. The alert message is called a _____.

Question 2

SNMPv1 and v2c transmit community strings in _____. SNMPv3 adds _____ and _____. Which version should be used on any production network and why?

Question 3

NetFlow exports _____ data from routers and switches — source/destination IP, port, protocol, byte count, and timestamps — without capturing the actual payload. Name two security use cases for NetFlow data.

Question 4

Syslog collects log messages from network devices to a central _____ server. Severity levels range from _____ (most critical, level 0) to _____ (least critical, level 7). Why is centralized logging more valuable than reading logs on each device individually?

Question 5

Interface utilization monitoring tracks the percentage of available bandwidth in use on each link. A link consistently running above _____ % utilization is considered congested and a candidate for upgrade or load balancing. What user symptom typically accompanies sustained high utilization?

Question 6

A network monitoring system uses _____ to determine if a device is reachable. If a device stops responding, the system generates an _____ to notify the administrator. What is the risk of alert fatigue in a monitoring system?

Question 7

Port mirroring (SPAN) copies traffic from one or more _____ to a designated _____ port connected to a monitoring device like Wireshark or an IDS sensor. What is the limitation of using a SPAN port for IDS compared to an inline IPS?

Question 8 — Real World Application

REAL WORLD: A school's internet link was saturated every day from 2–3 PM for three weeks before anyone noticed. No monitoring was in place. Describe a monitoring setup using two tools from this lesson that would have detected this within the first day, and what each tool would have shown.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/3-1-2-network-monitoring/3-1-2-network-monitoring-answer-key.pdf
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